

Joseph Walton

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Professional Summary

Senior AI and machine learning engineer with over 10 years of experience delivering production systems across regulated government, financial services, intellectual property and defence environments. Hands-on expertise in Python, Generative AI, RAG, agentic systems, LangChain, LangGraph and Azure AI services. Proven technical leader across architecture, implementation, evaluation, observability and multi-environment deployment, with delivered solutions saving approximately £2M annually. Strong advocate for secure, responsible AI who communicates effectively with engineering, architecture, business and international stakeholders.

Technical Skills

- **Generative AI:** Azure OpenAI, Azure AI Foundry, LangChain, LangGraph, RAG, agentic AI, LLM evaluation and observability
 - **Programming & Integration:** Python, SQL, Java, Azure Functions, service-oriented systems
 - **Search & Data:** Elasticsearch, vector search, HNSW, Apache Spark, Polars, Pandas, NumPy
 - **MLOps & CI/CD:** Azure ML Pipelines, Vertex AI Pipelines, MLflow, Git, Docker, multi-environment deployment
 - **Cloud Platforms:** Microsoft Azure, Google Cloud Platform, AWS
 - **Machine Learning:** Hugging Face, BERT, TensorFlow, PyTorch, Scikit-learn
 - **Security:** AI red teaming, adversarial testing, secure containerised deployment, responsible AI
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Professional Experience

IBM

Lead Data Scientist

June 2025 - Present | Contract role delivering AI solutions for central government

- Designed and delivered a production agentic data-processing solution with Python, Azure AI Foundry, Azure Functions and LangGraph, saving case workers two hours per day and approximately £2M annually.
- Rescued underperforming AI projects by introducing systematic evaluation and observability practices.
- Built a Claude Skill to streamline human-in-the-loop data ingestion for LLM evaluation, saving approximately one hour per evaluation case.
- Re-engineered a data-matching algorithm in Polars, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.

Tech Stack: Python, Azure AI Foundry, Azure Functions, LangGraph, Polars

Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

- Developed production Generative AI applications within a service-oriented architecture using Python, LangChain, Azure OpenAI and Azure Functions.
- Built a RAG agent over unstructured text to help users identify relevant Goods and Services terms.
- Implemented and tuned Elasticsearch HNSW vector search, balancing throughput and recall through systematic evaluation.
- Improved search and recommendation relevance metrics by 20% through data-driven experimentation and optimisation.
- Implemented an end-to-end MLOps strategy across multiple environments, reducing model deployment time to under one day.
- Conducted red-team exercises to assess Generative AI applications against adversarial attacks and guide robustness improvements.
- Set the product roadmap for AI applications, balancing opportunities, risks and delivery constraints.
- Advised technical and non-technical stakeholders across international intellectual property offices and contributed to cross-government AI policy and best practices.
- Replaced Azure Databricks ETL workloads with Polars pipelines, cutting compute costs by 50% and enabling zero-downtime deployments.

Tech Stack: Python, Azure OpenAI, LangChain, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

- Led cross-functional delivery of secure, production machine learning solutions using large-scale text and image data.
- Designed and implemented an MLOps framework on Google Cloud Platform, orchestrating training, evaluation, approval and deployment with Vertex AI Pipelines and reducing time to market to under one day.
- Developed and deployed a multilingual BERT model that detected intellectual property infringements across more than 90 languages.
- Built Spark pipelines for model training, batch inference and recommendation workloads.
- Mentored a software engineer through a transition into a machine learning role.

Tech Stack: Python, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

- Engineered a production NLP system in a secure, containerised environment, increasing enforced infringements by 10% per analyst across approximately 200 analysts.
- Built and deployed an image-similarity model that helped secure a major blue-chip contract representing 10% of annual recurring revenue.

Tech Stack: Python, TensorFlow, FastText, Google Cloud Platform, MLflow, Docker

Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

- Developed Python machine learning services and scalable data pipelines to integrate and process sensitive administrative datasets from multiple sources.
- Worked in a cross-functional team to develop a highly scalable backend for the UK Census 2021 data-collection system.

Tech Stack: Python, PySpark, Cloudera

Barclays

Data Scientist

September 2018 - October 2018 | Secondment

- Applied statistical and analytical methods to securely and ethically use payments data for regional economic indicators.
 - Continued work from the winning ONS x Barclays Hackathon team.
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Purple Secure Systems

Software Engineer

August 2014 - October 2016

- Developed Python and Java big-data processing systems for government and defence clients.

Tech Stack: Java, Python, PySpark, AngularJS

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honors & Awards

Winner, ONS x Barclays Data Hackathon

Used payments data to create more granular and timely estimates for Gross Value Added (GVA).